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Few-layer sp^2 carbon supported on Al_2O_3 as hybrid structure for ethylbenzene oxidative dehydrogenation

Jia Wang[a], Jiangyong Diao[a], Jiayun Zhang[a,b], Yajie Zhang[a,b], Hongyang Liu[a,*], Dang Sheng Su[a,*]

[a] Shenyang National Laboratory for Materials Science, Institute of Metal Research, Chinese Academy of Science, Shenyang, 110016, China
[b] School of Chemistry and Materials Science, University of Science and Technology of China, Hefei, 230026, China

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ABSTRACT

Hybrid materials consisting of few-layer sp^2 carbon supported on Al_2O_3 can potentially display comparable performance with nanodiamonds for the oxidative dehydrogenation of ethylbenzene. The hybrid structure could be easily prepared by fast coking method using ethanol as a precursor and γ-Al_2O_3 with high specific surface areas as a template. In this case, γ-Al_2O_3 plays the role of 3D support for depositing few-layer graphene-like structure. The hybrid structure exhibiting similar structure with nanodiamonds calcined at high temperature is further used as a metal-free catalyst for the oxidative dehydrogenation of ethylbenzene to styrene. The yield rate of styrene normalized by the specific area on Al_2O_3@C is 0.044 mmol m^{-2} h^{-1} (TOF = 4.13 h^{-1}) which is two times of that on nanodiamonds (0.020 mmol m^{-2} h^{-1}, TOF = 3.27 h^{-1}). Considering the low cost process and good catalytic performance, the hybrid nanostructure may be a promising candidate to be applied in the oxidative dehydrogenation of ethylbenzene.

1. Introduction

Carbon-based catalyst has attracted great interest in the past decades due to its unique properties for the direct dehydrogenation (DDH) and oxidative dehydrogenation (ODH) of hydrocarbons [1,2]. Among many kinds of nanocarbons (e.g. carbon nanofibers, mesoporous carbon), it is found that nanodiamonds (NDs) and onion-like carbons (OLCs) display a high catalytic activity in DDH/ODH of ethylbenzene (EB) to styrene (ST) [3–5]. The high activity of the NDs may be attributed to the presence of a unique sp^2-sp^3 carbon hybrid structure, which provides a partial delocalization of the electron density [6]. Presently, most of the commercially available NDs in powder form are produced by an explosion method followed by nitric/sulfuric acid purification, resulting in high price of NDs and a mass of waste acid [7]. Moreover, the surface of fresh NDs usually contains amorphous carbon [8]. In order to obtain sp^2-sp^3 hybrid NDs, it is necessary to treat NDs at high temperature, generally above 1100 °C [9]. Therefore, it is imperative to further explore available and low-cost alternatives with comparable catalytic performance.

Recently, the template (e.g. metal oxides) method have attracted considerable attention because the structure of as-prepared carbon materials could be controlled by tuning the size or morphology of the template [10,11]. After the removal of template (generally, acid treatment method), hollow carbon nanostructures with amorphous or graphitic structure will be obtained, which has attracted considerable attention owing to their potential applications in catalyst supports, and lithium-ion batteries [12–14]. For the template method, the core-shell structure is synthesized by coating a carbon precursor on a hard template core, followed by carbonization or directly by CVD method. It is available to obtain a hybrid structure consisting of few-layer sp^2 carbon supported on a hard template core by regulating the synthesis conditions, such as temperature, the amount and type of precursor.

Here in this paper, we report a facile and large-scale method to synthesize hybrid nanostructure consisting of γ-Al_2O_3 core wrapped with 2–5 layer graphitized carbon (Al_2O_3@C) by a CVD process at 800 °C using commercial γ-Al_2O_3 powder as a template. The as-prepared hybrid nanostructure is used as a catalyst for the ODH of EB and exhibits a comparable activity with NDs. The yield rate of styrene normalized by the specific area on Al_2O_3@C is 0.044 mmol m^{-2} h^{-1} (TOF = 4.13 h^{-1}) which is two times of that on NDs (0.020 mmol m^{-2} h^{-1}, TOF = 3.27 h^{-1}). Considering the low cost process and good catalytic performance, the hybrid nanostruc-

* Corresponding authors.
E-mail addresses: liuhy@imr.ac.cn (H. Liu), dssu@imr.ac.cn (D.S. Su).

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